VINICIUS C. LOPES

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EDUCATION

UNIVERSITY OF NOTRE DAME — Notre Dame, IN
Ph.D. Candidate in Computer Science and Engineering — **Expected**: May 2027
Dissertation: *Software Vulnerability Lifecycle: Detection, Validation and Remediation of Security-Critical Code Changes*

FEDERAL UNIVERSITY OF GOIAS (UFG) — Goiania, Brazil
M.Sc Computer Science — April 2022

FEDERAL UNIVERSITY OF GOIAS (UFG) — Goiania, Brazil
B.Sc. Software Engineering, National Scholarship — Dec, 2018

PUBLICATIONS

- **Vinicius Carvalho Lopes**, Matthew Zampino, Christian Garcia, Joanna C. S. Santos, and Adriana Sejfia. *On the Quality of Vulnerability-Contributing Commit Datasets: A Systematic Review and Exploit-Based Evaluation*. ESEM 2026 (19% acceptance rate), LIPIcs vol. 394, 46:1-46:21. doi:10.4230/LIPIcs.ESEM.2026.46
- Mohammed Latif Siddiq, Xinye Zhao, **Vinicius Carvalho Lopes**, Beatrice Casey, and Joanna C. S. Santos. *Security in the Age of AI Teammates: An Empirical Study of Agentic Pull Requests on GitHub*. Information and Software Technology, 2026, article 108333.
- Mohammed Latif Siddiq, Joanna C. S. Santos, Ridwanul Hasan Tanvir, Noshin Ulfat, Fahmid Al Rifat, and **Vinicius Carvalho Lopes**. *Using Large Language Models to Generate JUnit Tests: An Empirical Study*. International Conference on Evaluation and Assessment in Software Engineering (EASE), 2024.
- **Vinícius C. Lopes**, Cleiton Silva, Dayana Gonçalves, Roberto Oliveira, Renato Bulcão-Neto, Mohamad Kassab, and Valdemar V. Graciano-Neto. *A Systematic Mapping Study on IoT-Based Software Systems for Precision Agriculture*. International Journal of Computer Applications in Technology, 2022.

ACHIEVEMENTS AND AWARDS

- Tuition scholarship, including health insurance coverage and bimonthly stipends from the University of Notre Dame.
- Kylemore Abbey Writing Retreat award, granted by the University of Notre Dame College of Engineering to selected PhD students for a dissertation writing retreat at the Notre Dame Kylemore Abbey Global Centre in Ireland (2026)
- Student Volunteer, International Conference on Software Engineering (ICSE 2022), Pittsburgh, PA, USA
- NSF Student Travel Grant to attend the International Conference on Software Engineering (ICSE 2022), Pittsburgh, PA, USA
- CAPES scholarship (Coordination for the Improvement of Higher Education Personnel) granted by the Brazilian Government to work as a research assistant at the Federal University of Goias
- CNPq scholarship (National Council for Scientific and Technological Development) granted by the Brazilian Government to work as an undergraduate research assistant at the Federal University of Goias

RESEARCH EXPERIENCE

UNIVERSITY OF NOTRE DAME — Notre Dame, IN
Software Security Research Assistant — January 2022 - Now

- Conducted research on state-of-the-art methods for identifying bugs and vulnerabilities at commit time. We replicated datasets using Docker.
- Built a Docker-based exploit validation pipeline for CVEs across Apache Struts, Tomcat, ActiveMQ, and other projects; found a 79.2% false positive rate among 168 exploit-validated VCC labels from 6 existing datasets (3,105 samples), published at ESEM 2026.
- Developed a synthetic dataset of Vulnerability-Contributing Commits (VCCs) using Large Language Models (LLMs), code analysis techniques, and the WALA framework in Java and some Python scripts.
- Designed and evaluated machine learning models with novel security features to improve VCC identification.
- Leading an empirical study comparing security-related and non-security pull requests across multiple project metrics, including which CWEs are hardest to patch.
- Building a commit analyzer in Java, leveraging static code analysis to detect vulnerability-introducing changes and their locations in the codebase.

UNIVERSITY OF NOTRE DAME Notre Dame, IN
Research Mentor, Computing Skills Accelerator (CSA) Program Summer 2025

- Mentored two students with no research experience during the summer.
- Designed and scoped a project with clear goals, timelines, and milestones.
- Guided students in developing a synthetic dataset of Vulnerability-Contributing Commits (VCCs) using Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs).
- Supervised the preparation and presentation of their results at a research symposium poster session.

FEDERAL UNIVERSITY OF GOIAS Goiania, GO
Research Assistant 2020-2022

- Designed and implemented an IoT architecture using C/C++ and Arduino to monitor temperature and humidity and automate climate control in an aviary.
- Deployed the system on a production farm, ensuring reliable operation and collecting real-world performance data over one month.
- Developed a JavaScript web application providing farmers with real-time monitoring and remote access to environmental data.

Undergraduate Research Assistant 2014-2018

- Conducted research project on applying ubiquitous computing, software architecture, and computational intelligence to home patient monitoring.
- Presented it as a poster at CBIS 2016 (Brazilian Congress on Health Informatics).

PROFESSIONAL EXPERIENCE

MAGNA INTERNATIONAL Detroit, MI
Embedded Software Intern Summer 2023 - Summer 2023

- Developed and optimized an algorithm that enabled the transmission and reception of camera images in cars. We used two USB cameras and two SK-TDA4VM boards.
- Developed algorithms that would reduce the input delays from cameras and what is actually being shown in the display of a car.
- Used C/C++ to develop those algorithms. I worked with encoders VP9, H264 and H265.

SKILLS

Programming Skills: Java, C, C++, C#, JavaScript, Python, Ruby on Rails, Arduino, Kotlin
Operation Systems: Ubuntu, RedHat, Windows, MacOS
Database: MySQL, PostgreSQL
Technical Writing: Latex, Overleaf.
Platform: Docker.
Frameworks: Wala.
Version Control:Github
Languages: Portuguese (Native), English (Proficient), Spanish (Proficient).

LEADERSHIP & SERVICES

WARRIOR-SCHOLAR PROJECT Notre Dame, IN
Instructor (Veterans Program) Summer 2025

- Prepared and designed classes of Python for veterans
- Designed a project that the veterans would develop during the classes. The project involved the use of the Pandas library to work with data and then apply it to a simple ML model

UNIVERSITY OF NOTRE DAME Notre Dame, IN
Programming Paradigms Teaching Assistant August 2025 – Now

- Managed 50+ students, held office hours, and graded homework in Java and Kotlin.

UNIVERSITY OF NOTRE DAME Notre Dame, IN
Programming Paradigms Teaching Assistant January 2022 – December 2022

- Managed 140+ students, prepared teaching materials and lectured 2 classes on Java.

44TH INTERNATIONAL CONFERENCE ON SOFTWARE ENGINEERING (ICSE 2022) Pittsburg, PA
Volunteer May 2022

- Provided support in areas such as registration, event logistics, and presentation management, resulting in a seamless and efficient experience for participants.